



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This project focuses on the design and implementation of an autopilot system for the longitudinal dynamics of an aircraft, with the goal of achieving control and stability over vertical speed and pitch rate. The primary objectives include developing a vertical speed guidance loop, integrated with an inner pitch rate control loop, and an auto-throttle system to regulate airspeed. Through theoretical analysis and simulation, the system's performance was evaluated based on key criteria, including response time, stability, and robustness to external disturbances. A 6DOF nonlinear simulation was used to validate the control laws, demonstrating the autopilot's ability to track vertical speed and airspeed commands with minimal overshoot and fast settling times. The system's response to gust inputs was also assessed to evaluate its disturbance rejection capabilities. The results confirm that the proposed longitudinal autopilot design meets the specified performance requirements, providing a reliable solution for maintaining desired flight paths and stability in the longitudinal plane.

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1. Introduction

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2. Q1 – Vertical Speed Transfer Function

To assess the feasibility of using the aircraft elevator (δ_e) to directly control the vertical speed (v_s), the transfer function $G_{\delta_e}^{v_s}$ was determined. Where $C = [0 \ -1 \ 0 \ V_{trim}]$, the Transfer Function for v_s was found to be (2.1).

$$G_{\delta_e}^{v_s}(s) = \frac{8.57(s - 16.89)(+12.79)(s + 0.04162)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (2.1)$$

The stability of aircraft's control system can be determined by the location of its poles, shown in Figure 1. Poles can be seen on the right half of the s-plane, demonstrating instability.

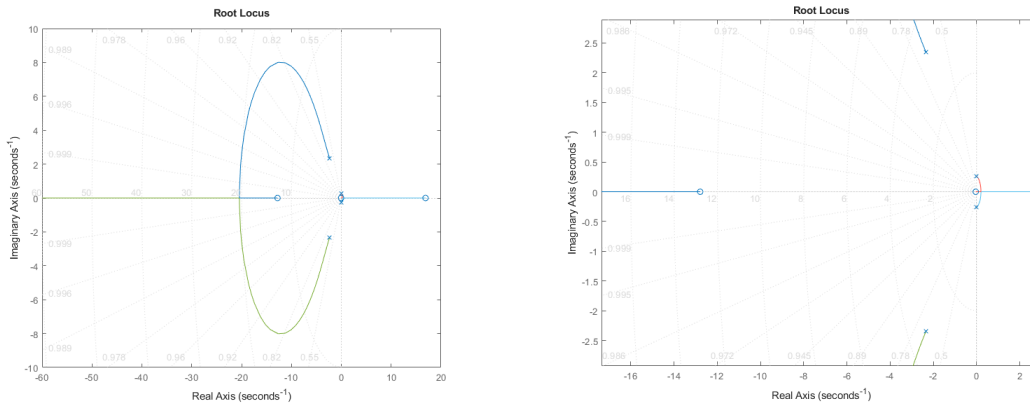


Figure 1: Root Locus of $G_{\delta_e}^{v_s}(s)$

Further, when analysing the Bode Plot of $G_{\delta_e}^{v_s}(s)$ in Figure 2, a resonant peak at a low frequency can be seen, indicating high sensitivity and possible instability. The phase response can be lagging at low frequencies. Thus, designing an autopilot using elevator to directly control vertical speed would be challenging due to the indirect relationship between elevator input and vertical speed output.

3. Q2 – Pitch Rate Autopilot

the following is for my understanding -

The primary function of the elevator is to control the pitch, which influences the angle of attack, which then in turn affects the lift produced by the wings. When the pitch angle is altered, there is a delayed response as the aircraft's trajectory adjusts to the new wing orientation. This is because the resultant lift force must change sufficiently to alter to the vertical component of the aircraft's velocity.

with no controller: settling time NaN, overshoot Inf,

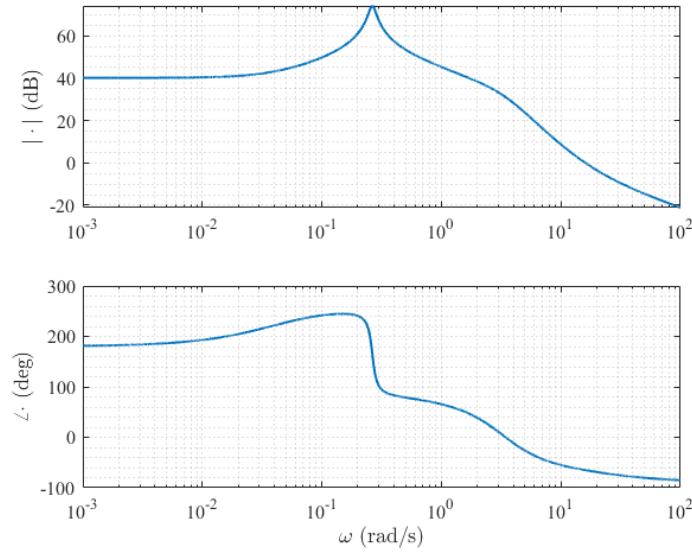


Figure 2: $G_{\delta_e}^{v_s}(s)$ Bode Plot

The Transfer function for elevator deflection to pitch rate q was found to be Equation 3.1, with root locus of that in Figure ???. The step response of the system was unstable, with 'Inf' overshoot and did not settle. Where $C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$,

$$G_{\delta_e}^q(s) = \frac{-26.397s(s + 1.767)(s + 0.08172)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (3.1)$$

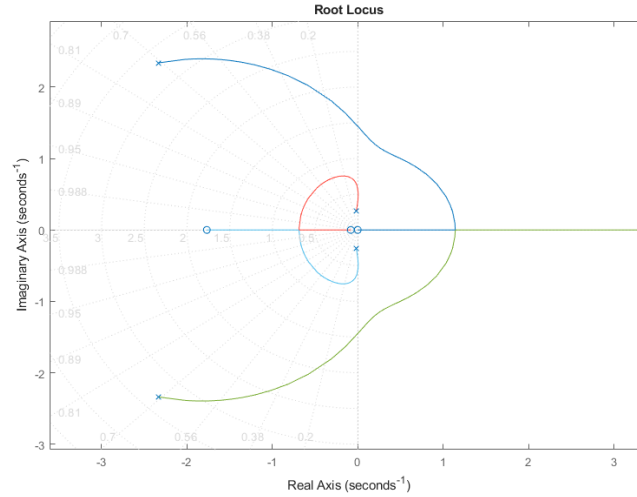


Figure 3: $G_{\delta_e}^q(s)$ Bode Plot

In order to design a controller to satisfy a bandwidth of $\omega_c = 2 - 10 \text{ rad/s}$, noise bandwidth of 20 rad/s, overshoot within 2% with settling time of around $3s$, it was determined that that system should have damping ratio (ζ) of 0.78 and Phase Margin of approximately 69.

A proportional integral derivative (PID) controller, with an additional pole, was implemented. The PID parameters were initially estimated and subsequently fine-tuned using the MATLAB 'Control System Designer' tool. Thus resulting in the controller in Equation 3.2.

$$C = \frac{-0.13015(s + 7.692)(s + 0.5263)}{s(s + 1.22e - 5)} \quad (3.2)$$

After applying the PID controller 3.2, the system had settling time of 2.7s and overshoot 8%.

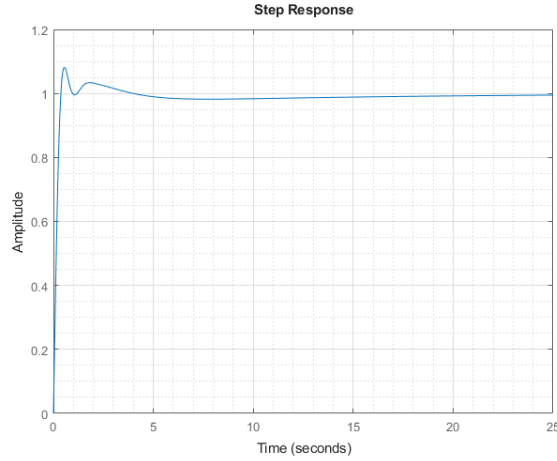


Figure 4: Step response of Controlled System

The Bode Plot in Figure 5 shows the system Phase Margin of 64.5 and frequency 5.74 rad/s, close to the desired requirements. While the designed pitch rate autopilot improved the pitch rate control performance, it did not fully meet all the specified criteria. The controller achieved desired performance in terms of settling time and Phase Margin, and effectively attenuates noise above 20 rad/s. However the response has 5% more overshoot than desired.

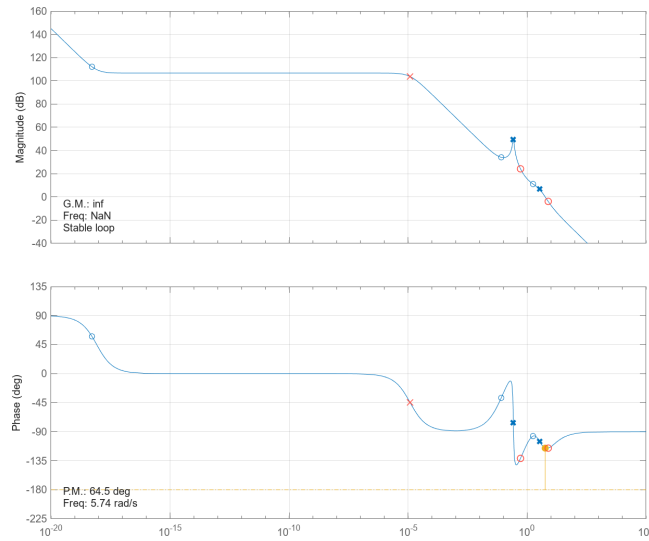


Figure 5: Bode Plot of Controlled System

4. Q3 – Vertical Speed Guidance Loop

The vertical speed guidance loop is designed to control the aircraft's ascent and descent rates by incorporating a pitch rate controller as an inner loop. This loop includes a compensator in the forward leg, which processes the vertical speed error to generate a pitch rate command. The pitch rate command then feeds into the inner loop to modulate the elevator input, allowing the system to adjust vertical speed in response to deviations from the desired reference.

4.1. Loop Topology

The complete loop topology of the vertical speed guidance loop is shown in Figure 6. The compensator transforms the vertical speed error ($vs_{ref} - vs$) into a pitch rate command. This command is then processed within the inner loop section ($de2q$), where the transfer functions G_1 and G_2 are used in the pitch rate controller to compute the required elevator deflection (δ_e) to achieve the target pitch rate.

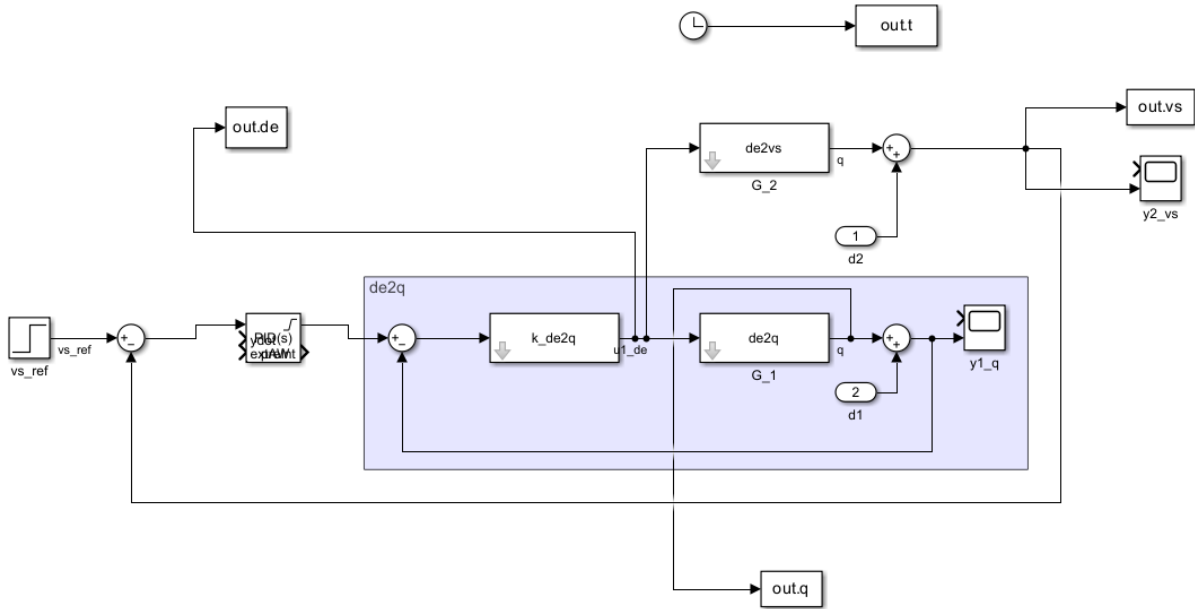


Figure 6: Vertical Speed Guidance Loop Topology

4.2. Aircraft Response to Vertical Speed Command

The vertical speed response, in Figure 7, demonstrates the ability for the system's response to a 500 ft/min (2.54 m/s) vertical speed step input command.

The vertical speed response, shown in Figure 2, demonstrates the system's performance in tracking a 500 ft/min (approximately 2.54 m/s) vertical speed command step input. The response shows a rapid increase in vertical speed, reaching close to the desired 2.5 m/s target within the first 2 seconds. There is a slight overshoot beyond the target speed, followed by a gradual settling towards

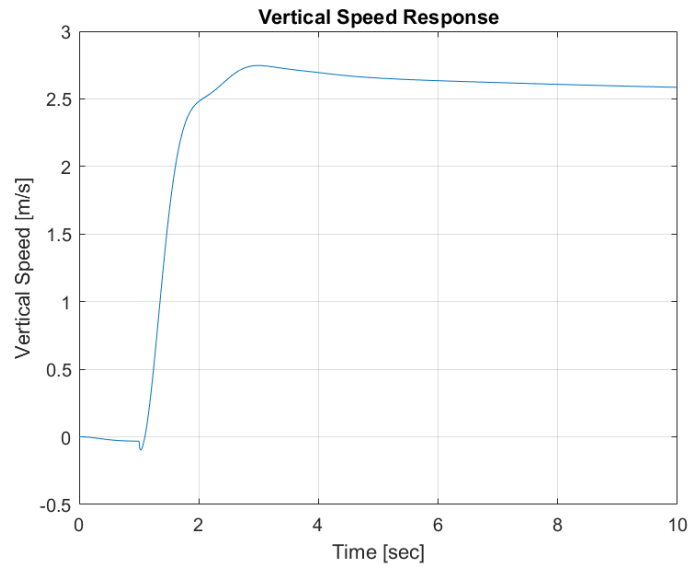


Figure 7: Vertical Speed Response

the reference value, achieving stability within approximately 10 seconds. This response indicates that the system meets the rise time and settling time requirements, effectively balancing speed of response with controlled damping. The smooth convergence to the command confirms the system's capability to achieve accurate vertical speed tracking.

4.3. Pitch Rate Response

The pitch rate response, shown in Figure 8, reflects the performance of the inner pitch rate controller.

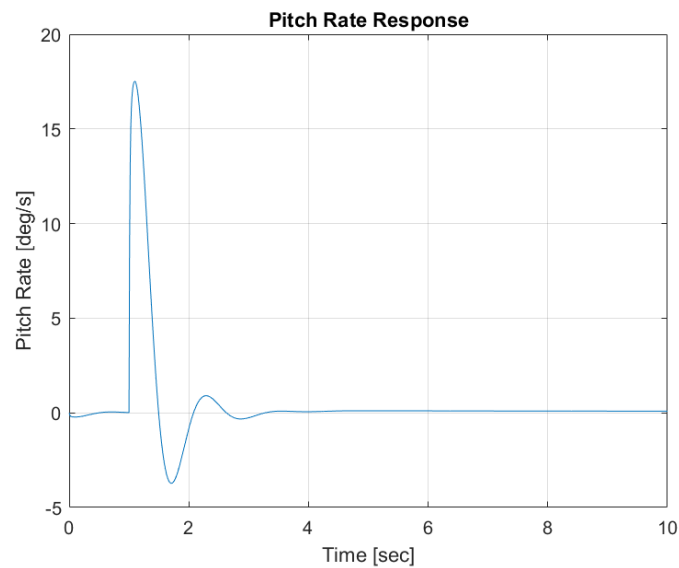


Figure 8: Pitch Rate Response

The response shows a peak reaching 16deg/s , followed by a quick correction with minor oscillations before settling to zero. This peak and correction indicate the controller's rapid adjustment to the pitch rate required to meet the vertical speed command of 500ft/min (2.54m/s), as outlined in Section 4.2. These damped oscillations demonstrate the pitch rate controller's stability and responsiveness.

4.4. Elevator Activity

Figure 9 displays the control activity of the elevator in response to the inputted vertical speed command.

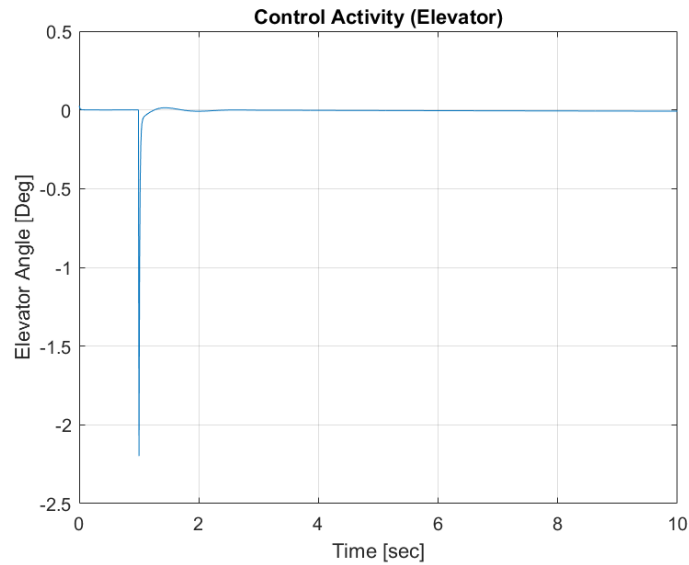


Figure 9: Elevator Control Activity

The elevator initially deflects sharply to approximately -2.3degrees as the controller adjusts to achieve vertical speed command, followed by the elevator deflection stabilizing back to zero, demonstrating its minimal effort required once the target vertical speed is reached.

5. Q4 – Design Auto-Throttle Control System

In this section an auto-throttle design was attempted, wherein errors in airspeed would determine throttle command.

5.1. MIMO Loop Topology

The auto-throttle system was to be achieved with the MIMO loop topology shown in Figure 10.

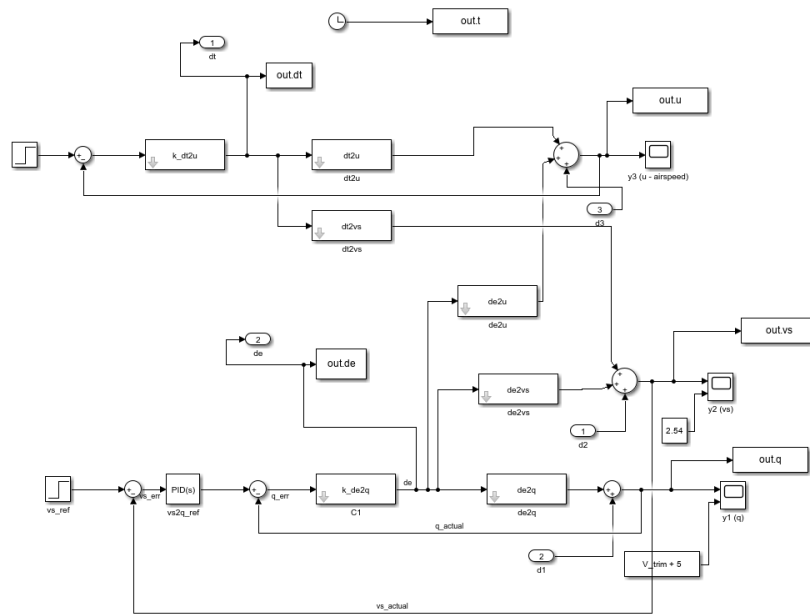


Figure 10: MIMO Simulink Loop for Auto-Throttle Design

5.2. Guidance Loop Topology Specifications

This loop was designed to the same specifications as the vertical speed guidance loop, with a vertical speed step input of 500 ft/min. The other constraints could not be implemented however due to difficulties in controlling the system. In addition to the vertical speed input there was also an input for u (airspeed).

5.3. Primary Control Effects

The primary control effects, namely the transfer function of throttle to airspeed ($(dt2u)$), and elevator to vertical speed ($de2vs$), are shown in Figures 11 and 12.

These effects tend to follow a more sinusoidal pattern, reaching similar magnitudes over the simulation period. This is quite different from the responses found from solely the vertical speed control loop (in

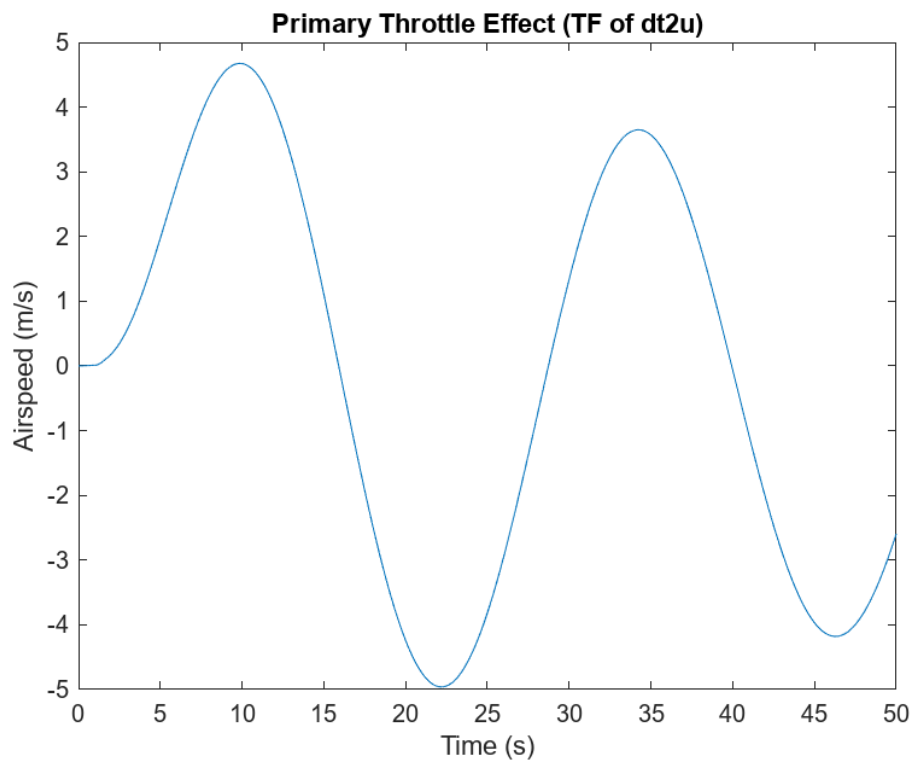


Figure 11: Primary Control Effect: Throttle to Airspeed

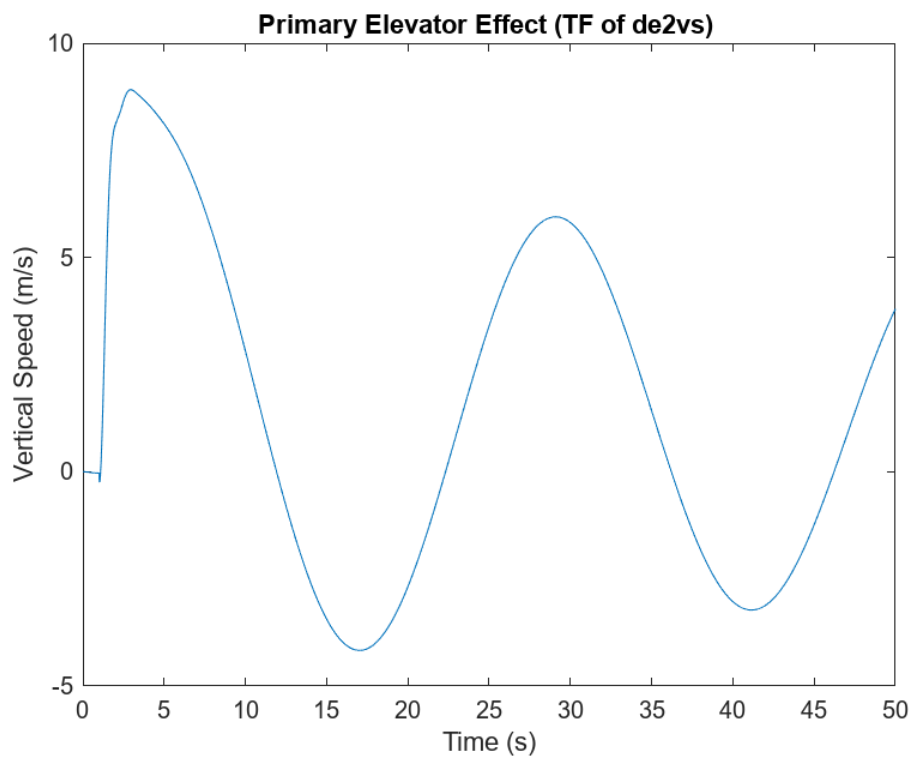


Figure 12: Primary Control Effect: Elevator to Vertical Speed

4.1). This indicates the the controls have more of an effect on the outputs over time, although because of the increased complexity of the system it is more difficult to tune, and does not reach a stable point (like the outputs of the vertical speed control loop). This throttle is more effective in making airspeed changes would be due to its direct affect on this output, coupled with the effects from the elevator control (as opposed to elevator control alone).

The overall output of vertical speed and pitch rate (from both elevator and throttle control) however show a more similar trend to the vertical speed control loop, shown in Figures 13 and 14, albeit with a longer settling time (i.e., not being controlled in the simulation time).

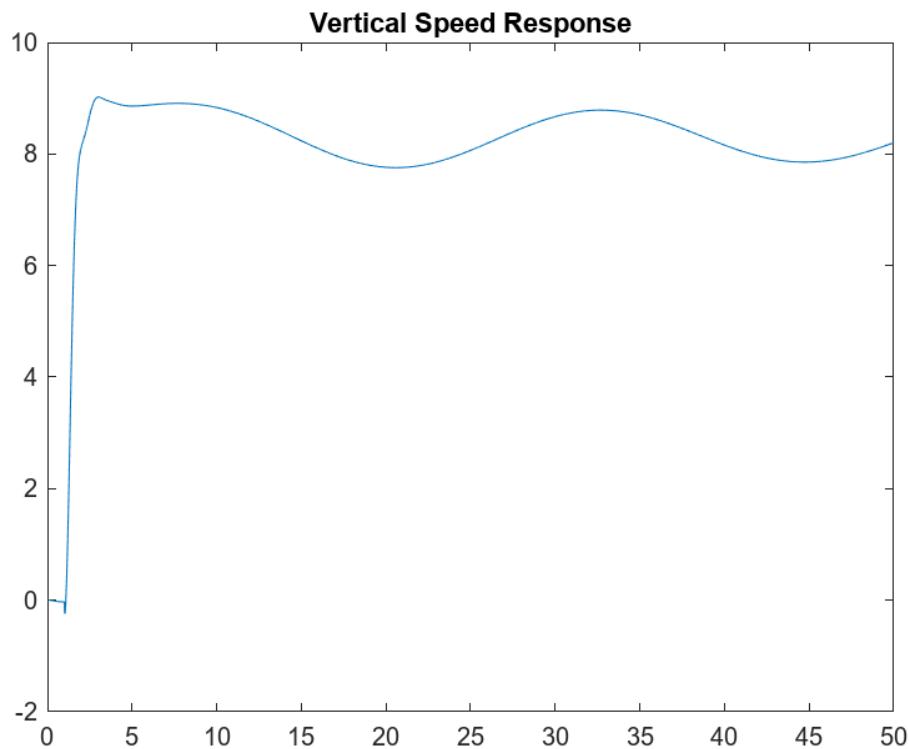


Figure 13: Vertical Speed Response (Auto Throttle System)

5.4. Cross-coupling Control Effects

The cross-coupling control effects, namely the transfer function of throttle to vertical speed ($(dt2vs)$), and elevator to airspeed ($(de2u)$), are shown in 15 and 16.

Again, compared to only the vertical speed control loop, these reponses are sinusoidal, and do not stabilise in the timeframe - showing the complexities of cross looping.

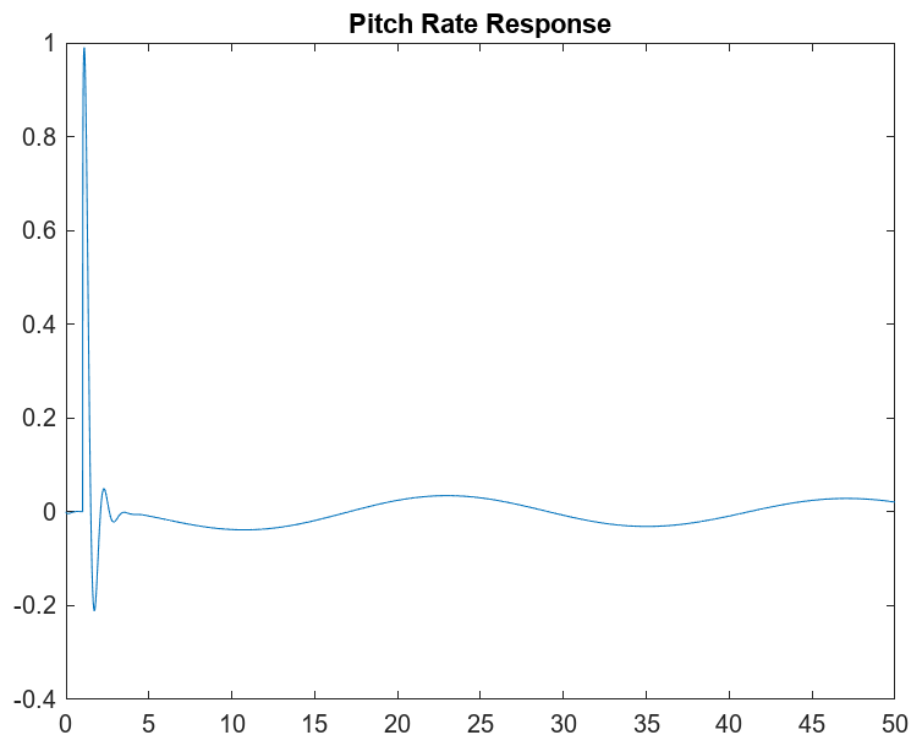


Figure 14: Pitch Rate Response (Auto Throttle System)

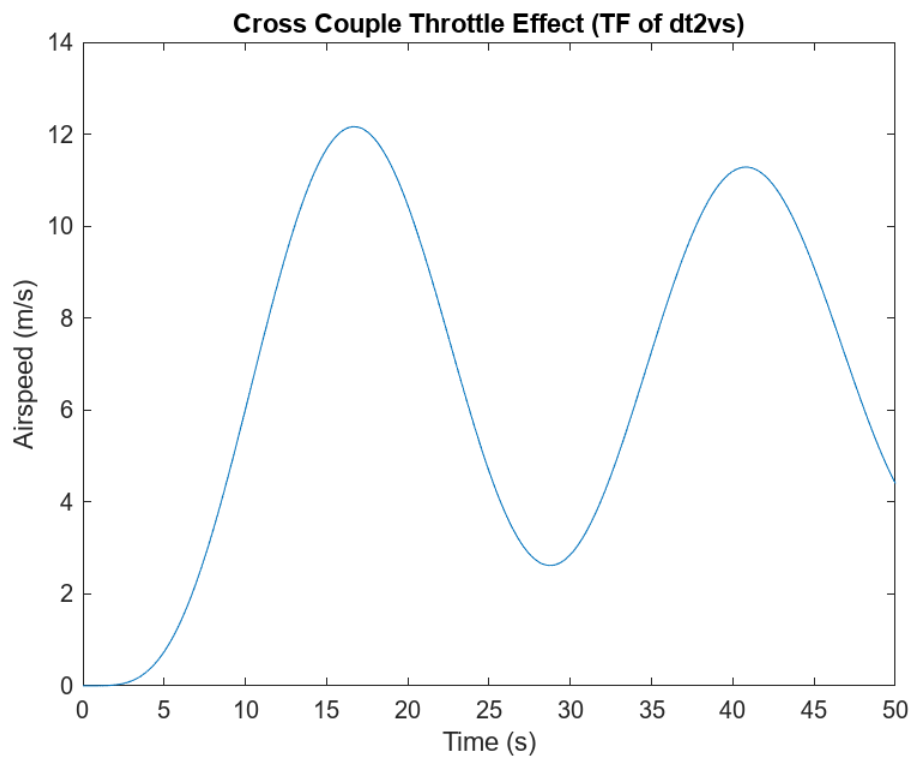


Figure 15: Cross Couple Response (Throttle to Vertical Speed)

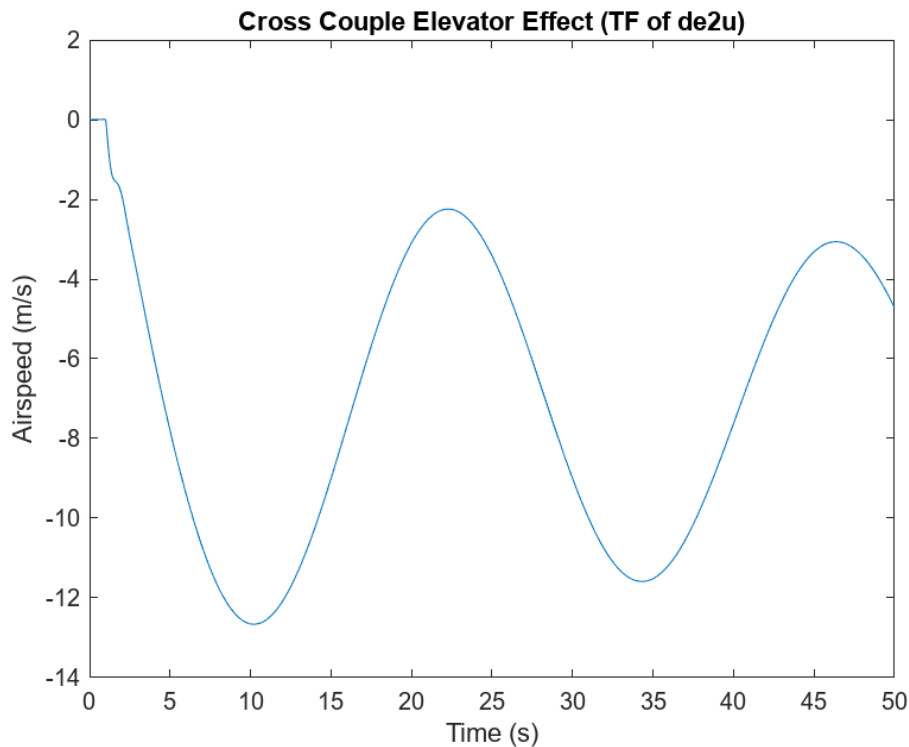


Figure 16: Cross Couple Response (Elevator to Airspeed)

5.5. Elevator and Throttle Response

5.6. Auto-Throttle Effectiveness

6. Final Design

6.1. Q5 – Implementation of the control and guidance laws

A six degree of freedom (DOF) simulation includes all translational and rotational motions, capturing complex interactions and nonlinearities, unlike the transfer function models which simplify the behaviour of the system around a specific point.

In Figure 17 and Figure 18, the MIMO controller was represented as a state-space system.

The linear results of Figure 17 are as expected, !!REF:same as q4!!. The coupling effects are highlighted when analysing the individual state responses in state-space form.

The nonlinear responses in Figure 18 appear more damped and exhibit less sensitivity to inputs, suggesting that the control law might be effectively compensating for changes. However, it is more plausible that this behavior is due to an incorrect implementation of the state-space model of the system. The variables u , w , q , and θ all show responses with reduced amplitude. While these differences could be attributed to nonlinear dynamics not accounted for in the linear model, they are more likely the result of inaccuracies in the model implementation.

6.2. Q6 – Gust Effectiveness

Disturbances are able to modelled in the six DOF nonlinear simulation.

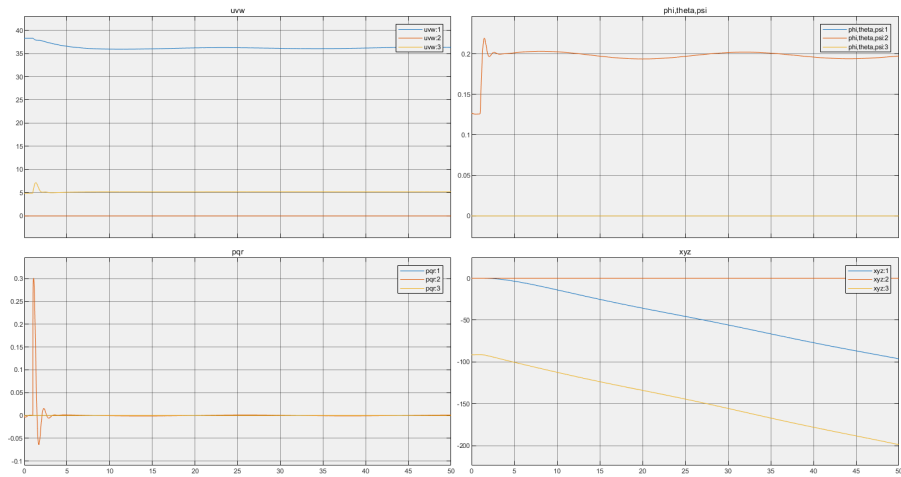


Figure 17: 6 DOF State-Space System Linear Response

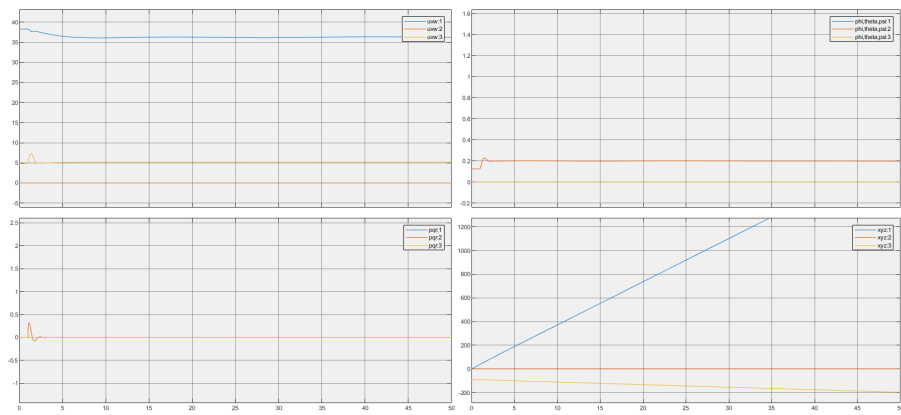


Figure 18: 6 DOF State-Space System Nonlinear Response

6.3. Q7 – Commanded Large Vertical Speed

could be dealt with by implementing filters in the control system to smooth out the control commands
more advanced control system - model predictive control

7. Conclusion

References

A. Example of a Table